Hafiza Kanwal Janjua

kanwal.janjua99@gmail.com
https://orcid.org/0000-0002-6655-6015
www.linkedin.com/in/hafiza-kanwal-janjua-84700995
+34604841070

PROFESSIONAL SUMMARY

Research Engineer at Barcelona Supercomputing Center (BSC-CNS) and PhD researcher specializing in machine learning and scientific computing for dynamic, multi-variable systems. I build data-driven models, simulation-based pipelines, and design constrained optimization/control methods validated through SLA/KPI instrumentation and rigorous benchmarking. I implement reproducible research in Python (Linux/Git) using TensorFlow/PyTorch, and I have a strong publication record and international collaboration experience.

TECHNICAL SKILLS

- **Programming:** Python
- **AI/ML:** Deep Learning (DL), Reinforcement Learning (RL), Machine Learning (ML)
- **Tools:** Linux, Git
- **Cloud/Orchestration:** Kubernetes, OpenStack

KEY RESEARCH AREAS & INTERESTS

- AI/ML for Networks (Optimization & Control)
- Agentic AI for Network & Cloud Operations
- Kubernetes & Cloud-Native Orchestration
- Autonomous Resource Management (Scheduling/Autoscaling)
- RAN Optimization & Radio Resource Management (RRM) using AI/ML

RESEARCH COLLABORATIONS (10Months)

- **Research Apprenticeship (PhD Secondment)**
  Telefónica, Spain | 3 months
  Conducted applied R&D on AI-enabled network orchestration/automation aligned with my PhD work in SDN/MEC and resource management.
- **Research Apprenticeship (PhD Secondment)**
  Aarhus University, Denmark | 2 months
  Investigated data-driven and ML-based optimization for next-generation networks, supporting my PhD research.
- **Research Apprenticeship (PhD Secondment)**
  Chocolate Cloud, Denmark | 5 months
  Developed and validated cloud-native/edge orchestration prototypes integrating monitoring and automation workflows relevant to my PhD research.

REFERENCES

- **Ignacio De Miguel Jiménez | Ph.D Supervisor**
  Professor, University of Valladolid (UVa), Spain
  Email: ignacio.demiguel@uva.es
- **Elli Kartsakli | Manager**
  Barcelona Supercomputing Center (BSC), Barcelona, Spain
  Email: elli.kartsakli@bsc.es

PERSONAL COMPETENCIES

- Collaborative team player with the ability to work independently when needed.
- Strong interpersonal, written, and verbal communication skills.
- Thrive in fast-paced, high-demand environments and adapt quickly to change.
- Proven ability to prioritize multiple deadlines while maintaining high quality and attention to detail.

LANGUAGES

- English – C2 Level
- Spanish – A2 Level
- Arabic – A2 Level

EDUCATION

- **P.HD. Telecommunication and Information Engineering** | Sep-2021- Mar 2026
  PhD Fellowship awarded under Marie Skłodowska-Curie Actions (MSCA), GA nº: 953442.
  University of Valladolid (UVa), Spain
  Thesis: Cognitive orchestrator of MEC and network resources
- **MS Information Security** | 2018-2020 | 3.59/4.00 CGPA
  COMSATS University Islamabad (CUI), Pakistan.
  Thesis: Privacy-Aware Log-Preservation Architecture in Fog-Enabled-Cloud Using Holochain and Containerization Technologies
- **BS Software Engineering** | 2012-2016 | 3.44/4.00 CGPA
  National University of Modern Languages (NUML), Pakistan.
  Thesis: Enhanced Secure Mechanism for Virtual Machine Migration in Clouds

WORK EXPERIENCE

- **Research Engineer**
  Barcelona Super Computing Center (BSC-CNS) | Jun. 2024 - Present
  Work Responsibilities contributed to projects (ASCENDER / EXTRACT / VERGE)
  - Conduct R2-level R&D on AI/ML-driven radio resource management (RRM) for beyond-5G/6G, including data-driven modelling, system simulation, and wireless network optimization under Software-Level-Agreement (SLA) and (Key Performance Indicator) KPI constraints.
  - Designed and evaluated quantum and classical RL/DL strategies for RAN control (e.g., bandwidth/PRB allocation, slice-aware policies) including traffic, mobility/channel modelling, KPI tracking (QoE, latency, throughput, SE/EE), and training/inference.
  - Implemented and integrated research software in Python using TensorFlow/PyTorch for training/inference, with Git/Linux-based reproducibility, and ML/RL controllers with orchestration and monitoring components.
  - Contribute to international & national R&D projects (Ascender, Verge, & Extract, And Proximity), producing technical reports, experiment documentation, and disseminate results through peer-reviewed papers, conference/workshop presentations, and internal/external technical talks.
  - Collaborate with multi-partner teams and stakeholders, translating research goals into test plans, benchmarks, and measurable outcomes, while delivering under project deadlines.
- **Research Fellow | MSCA-Early-Stage Researcher (R1)**
  Universidad de Valladolid (UVa), Spain | Sep. 2021- May 2024
  Work Responsibilities contributed to IoTalentum MSCA Project
  - Performed doctoral research on cognitive orchestration for SDN/MEC and RAN systems, including network slicing, service-aware resource allocation, and end-to-end optimization.
  - Built research-grade simulation and experimental frameworks (e.g., traffic modelling, KPI instrumentation, RL training/evaluation loops) to study orchestration policies under realistic constraints.
  - Designed and evaluated AI/ML-based decision-making approaches (e.g., RL agents) for resource management.
  - Produced peer-reviewed publications, research deliverables, and technical documentation aligned with project objectives and milestones.
  - Actively participated in research meetings, consortium activities, and scientific events, communicating results to both technical and non-technical audiences.
- **Research Assistant**
  COMSATS University Islamabad (CUI), Pakistan | Jan. 2019 – Sep. 2019
  Work Responsibilities contributed to CTIP Project
  - Supported the CTIP cyber threat intelligence R&D by building data pipelines and features, developing/evaluating ML-based detection with controlled experiments, and automating dataset preparation, runs, and reporting for reproducible security analytics.
  - Collaborated with supervisors and lab teams on requirements, experiment planning, and research documentation, and produced thesis/technical reports covering methodology, results, and limitations.
  - Contributed to the broader lab research direction around cloud/SDN, networking, and AI-assisted security, supporting demos, literature surveys, and experiment replication.
- **IT Engineer**
  Hum News Network (HNN), Pakistan | Jan. 2017 – Aug. 2018
  Work Responsibilities:
  - Managed day-to-day IT operations and end-user support (systems & software troubleshooting) to ensure uninterrupted business continuity.
  - Installed, configured, and maintained Windows workstations and enterprise tools, including updates, backups, and security hardening.
  - Supported and monitored LAN/Wi-Fi/network connectivity (basic routing/switching, issue resolution) and coordinated vendors for rapid fixes.

PUBLISHED & UPCOMING LIST OF PUBLICATIONS

[1]. **H. K. Janjua,** I. de Miguel, R. J. Durán Barroso, O. González de Dios, J. C. Aguado, N. Merayo, P. Fernández, and R. M. Lorenzo, "Efficient optimization of actor-critic learning for constrained resource orchestration in RAN with network slicing," in Proc. 26th Conf. Innovation in Clouds, Internet and Networks (ICIN), 2023, pp. 100–104, doi: https://doi.org/10.1109/ICIN56760.2023.10073489

[2]. **H. K. Janjua,** I. de Miguel, R. J. Durán Barroso, M. Masoumi, S. Hosseini, J. C. Aguado, N. Merayo, and E. J. Abril, "A framework for next generation cloud-native SDN cognitive resource orchestrator for IoTs (NG2CRO)," in Distributed Computing and Artificial Intelligence, Special Sessions I (DCAI 2023), Lecture Notes in Networks and Systems. Springer, 2023, pp. 399–407, doi: https://doi.org/10.1007/978-3-031-38318-2_39

[3]. M. Masoumi, I. de Miguel, R. J. Durán Barroso, S. Hosseini, **H. K. Janjua,** N. Merayo, J. C. Aguado, and R. M. Lorenzo, "Efficient protected VNF placement and MEC location selection for dynamic service provisioning in 5G networks," in Distributed Computing and Artificial Intelligence, Special Sessions I (DCAI 2023), Lecture Notes in Networks and Systems. Springer, 2023, pp. 448–456, doi: https://doi.org/10.1007/978-3-031-38318-2_44

[4]. S. Hosseini, I. de Miguel, R. J. Durán Barroso, N. Merayo, O. González de Dios, M. Masoumi, **H. K. Janjua,** J. C. Aguado, and P. Fernández, "Multi-band elastic optical networks: Overview and recent contributions from the IoTalentum project," in Distributed Computing and Artificial Intelligence, Special Sessions I (DCAI 2023), Lecture Notes in Networks and Systems. Springer, 2023, pp. 467–472, doi: https://doi.org/10.1007/978-3-031-38318-2_46

[5]. **H. K. Janjua,** M. A. Shah, A. Almogren, H. A. Khattak, C. Maple, and I. U. Din, "Proactive forensics in IoT: Privacy-aware log-preservation architecture in fog-enabled-cloud using holochain and containerization technologies," Electronics, vol. 9, no. 7, Art. no. 1172, 2020, doi: https://doi.org/10.3390/electronics9071172

[6]. **H. K. Janjua,** and W. Ali, "Enhanced secure mechanism for virtual machine migration in clouds," in Proc. Int. Conf. Frontiers of Information Technology (FIT), pp. 135–140, doi: https://doi.org/10.1109/FIT.2018.0003

[7]. **H. K. Janjua,** E. Kartsakli, and E. Quiñones, "Empirical analysis of intent-driven-orchestration in cognitive-RAN," IEEE Open Journal of the Communications Society (OJCOMS), accepted for publication, Dec 2025.

[8]. **H. K. Janjua,** I. de Miguel, N. Merayo, J. C. Aguado, E. J. Abril, and R. J. Durán Barroso, "Bridging 5G realities to 6G visions: A tutorial and a cognitive end-to-end framework for 6G orchestration and management," IEEE Open Journal of the Communications Society (OJCOMS), accepted for publication, Dec 2025.

[9]. **H. K. Janjua,** E. Kartsakli, and E. Quiñones, "MEC offloading meets quantum reinforcement learning," Proc. 29th Conf. Innovation in Clouds, Internet and Networks (ICIN), submitted, Dec 2025.

[10]. **H. K. Janjua,** I. de Miguel, N. Merayo, J. C. Aguado, E. J. Abril, and R. J. Durán Barroso, "Quantum deep learning model for 5G traffic classification and prediction," IEEE Access, submitted, Nov 2025.

Proof-of-Concepts (PoCs) Developed Related to Publications

- **Intent-Driven Orchestration in Cognitive-RAN using AI [1,7]**
  - Built an empirical framework mapping intent → policies → closed-loop actions, instrumented multi-KPI evaluation (e.g., QoE, latency, throughput, and Spectrum Efficiency), and performed rigorous experimental analysis across scenarios to validate stability, adaptability, and orchestration gains, against conventional RL/heuristic strategies.
- **MEC Offloading Meets Quantum Reinforcement Learning [9]**
  - Designed a quantum + classical RL decision loop for offloading actions (state → policy → allocation/offload), integrating MEC KPIs (delay/compute/utilization) and benchmarking methodology to compare quantum-RL behavior against conventional RL/heuristic strategies.
- **Quantum Deep Learning for 5G Traffic Classification & Prediction [10]**
  - Implemented a quantum-enhanced model pipeline (feature preparation → quantum encoding/learning → multi-class inference), evaluated with standard ML metrics (accuracy, confusion matrix-style validation), and positioned the model for integration into AI-driven traffic analytics and RAN/core automation loops.
- **Agentic AI–Powered Kubernetes Scheduling with Reinforcement Learning (Manuscript / In Preparation)**
  - Developed an agentic AI orchestration layer that combines reinforcement learning with Kubernetes scheduling to autonomously optimize placement and resource utilization for cloud-native microservices.
  - Implemented an RL-driven custom scheduling workflow (state from cluster telemetry → action as node/pod placement), integrated metrics pipelines (e.g., CPU/memory/network, latency, queueing/load) via monitoring, and packaged the system using cloud-native components (Kubernetes APIs, controller/scheduler hooks, Dockerized services, Git/Linux).
- **5G-to-6G Tutorial + Cognitive End-to-End Orchestration Framework [2, 8]**
  - Produced a tutorial-grade, end-to-end cognitive orchestration framework bridging real 5G constraints to 6G orchestration and management vision.
  - Defined a layered architecture spanning RAN/MEC/Core/Transport, formalized orchestration workflows (telemetry, decision, enforcement, assurance), and provided a system-level blueprint aligned with modern network automation concepts (closed-loop control, policy enforcement, KPI-driven optimization).